

PROJECT PLAN

AI Workforce OS

*An Enterprise Multi-Agent Platform for Business Operations***Course:** Web Services & Service-Oriented Architecture (IT644), Autumn 2026**Deliverable 2:** Project Plan (due 9 August 2026) — task timeline and milestones for the team and for individual members, team task assignments, tools, timeline chart and task board**Team:** 8 members · **Planning tool:** Trello · **Committed effort:** 1712 person-hours against a 1600-hour benchmark

1. Response to the Project Review Feedback

Observation raised	How this plan responds
1. Ideas approved; novelty not uniform in every case	Section 2 states the differentiator explicitly: not a single assistant, but a governed multi-agent workforce with shared memory, cited enterprise retrieval, least-privilege tool execution, human approval gates and a complete audit trail.
2. Overall project scope was not fully quantifiable	Section 3 replaces prose scope with counted, named artifacts, separated into committed scope and stretch scope.
3. Benchmark of 200 person-hours per member, 1600 in total, to be matched against task breakups and work assignment	Sections 4 to 8 cost every work package. The committed plan totals 1712 hours; the lowest individual total is 208 hours, so every member clears the 200-hour minimum from committed work alone.
4. Concerns to be addressed over the next two stages	This Project Plan is the first of those two stages. The Requirements Specification due 16 August will carry effort-costed user stories and acceptance criteria at the same granularity.

2. Scope Quantification

Committed scope is what the 1712 hours buy. **Stretch scope** is stated above that total and is dropped first, so dropping it cannot take any member below the 200-hour benchmark. Reductions against the approved proposal are declared here rather than left implicit.

Artifact	Committed	Stretch (stated above the committed total)
Backend services	7 (gateway/auth, organization, orchestrator, shared memory, knowledge/RAG, MCP integration, analytics)	—
REST endpoints	48, documented in OpenAPI	—
AI employee agents	6 core: HR, Engineering Manager, Developer, Research, Marketing, Customer Support	4 extended: Analytics, Product Manager, Sales, Finance (proposal listed 8 extended; 4 deferred to future work)
MCP server integrations	6: Gmail, Google Calendar, Slack, GitHub, Jira, Google Drive	4: Instagram, LinkedIn, CRM, AWS/Kubernetes
Knowledge connectors	6: Drive, Notion, Confluence, GitHub wiki, PDF, DOCX	—
UI screens	14 named screens	8 further screens
Data stores	3: PostgreSQL, Redis, Qdrant	Neo4j knowledge graph
Database tables	18	—
Event topics	6	—
Automated tests	approximately 150	—
Containers and manifests	8 images with Kubernetes manifests	—

Stretch scope is costed at 188 hours (4 extended agents 56, Neo4j 28, 4 further MCP servers 64, 8 further screens 40). Full ambition is therefore 1900 hours; the plan commits to 1712.

3. Estimating Basis

Hours are built up from per-artifact rates, not apportioned from the benchmark; the rates reproduce the Section 4 module totals to within normal estimating tolerance.

Unit of work	Rate	Unit of work	Rate
REST endpoint, including tests	1.5 h	Knowledge connector	8 h

Unit of work	Rate	Unit of work	Rate
UI screen, including state handling	6 h	Service scaffold with CI	10 h
MCP server, including OAuth	18 h	Module integration-test suite	8 h
AI agent: persona, tools, evaluation	16 h	Design or specification document	10-26 h

Cross-check: 6 committed MCP servers at 18 hours is 108 hours, against Ismail Mansuri's 136 build hours less approximately 28 hours for the shared permission-scope and approval-hook layer.

4. Effort Model

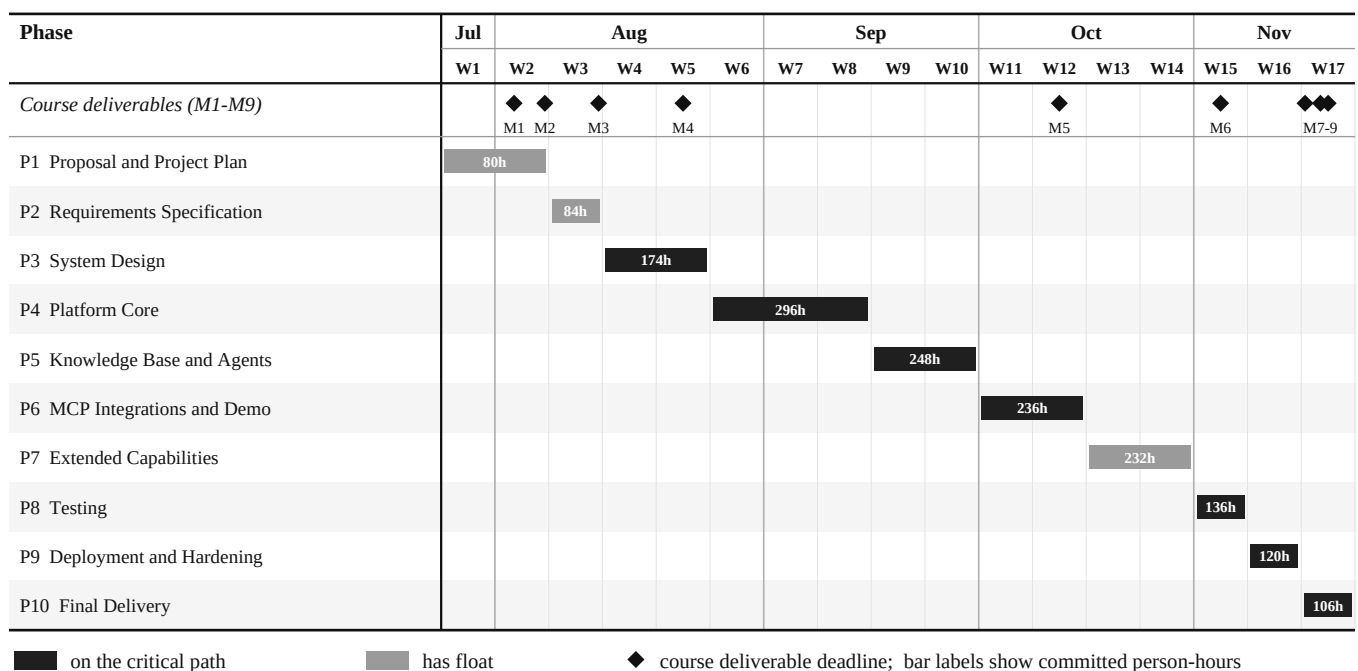
Rows are members, columns are the ten phases, cells are committed person-hours. Totals are the natural sum of the work packages in Section 8 and are deliberately not identical, because modules differ in size.

Member	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	Total	vs 200
Aum Parmar	18	12	26	26	22	28	24	20	16	16	208	+8
Love Shah	8	10	22	38	34	28	32	16	14	12	214	+14
Panthil Shah	8	10	22	26	36	36	32	16	12	12	210	+10
Fahim Saiyad	8	10	22	38	38	26	30	16	12	12	212	+12
Ismail Mansuri	8	10	22	44	36	32	24	18	12	12	218	+18
Yash Doshi	8	12	22	34	32	32	30	14	16	16	216	+16
Param Patel	8	10	18	56	26	26	28	16	18	12	218	+18
Afif Momin	14	10	20	34	24	28	32	20	20	14	216	+16
Phase total	80	84	174	296	248	236	232	136	120	106	1712	+112
Intensity, h/week/person	5.0	10.5	10.9	12.3	15.5	14.8	14.5	17.0	15.0	13.2	12.6	

Every member clears the benchmark from committed work alone (lowest total 208 hours). The team commits 1712 hours against the 1600-hour benchmark, a margin of 112 hours, or 7 per cent, which is the plan's contingency reserve. Average load is 12.6 hours per member per week; the highest single-week load is 20 hours and occurs only in the dedicated testing week.

5. Timeline Chart

The chart maps the ten phases onto the seventeen course weeks (W1 begins Monday 27 July 2026; W17 contains four working days). Bar labels give committed person-hours. Solid bars lie on the critical path defined in Section 10; lighter bars carry float. Diamonds mark the course deliverable deadlines of Section 6.



Phase schedule, effort and ownership

Phase	Weeks	Dates	Hours	Cards	Lead	Deputy
P1 Proposal and Project Plan	W1-2	27 Jul-9 Aug	80	13	Aum	Afif
P2 Requirements Specification	W3	10-16 Aug	84	14	Aum	Param
P3 System Design	W4-5	17-30 Aug	174	29	Aum	Yash
P4 Platform Core	W6-8	31 Aug-20 Sep	296	49	Param	Love
P5 Knowledge Base and First Agents	W9-10	21 Sep-4 Oct	248	41	Fahim	Panthil
P6 MCP Integrations and Mid-Semester Demo	W11-12	5-18 Oct	236	39	Ismail	Yash
P7 Extended Capabilities and Hardening	W13-14	19 Oct-1 Nov	232	39	Love	Panthil
P8 Testing	W15	2-8 Nov	136	23	Afif	Param
P9 Deployment and Hardening	W16	9-15 Nov	120	20	Afif	Ismail
P10 Final Delivery	W17	16-19 Nov	106	18	Aum	Yash

6. Milestones

Team milestones are the course deliverable deadlines. Every one is covered by a phase in Section 5 and by costed tasks in Section 8.

#	Date	Course deliverable
M1	Wed 5 Aug	Project Proposal Document
M2	Sun 9 Aug	Project Plan: timeline chart and task board (this document)
M3	Sun 16 Aug	Requirements Specification (BRD), effort-costed
M4	Thu 27 Aug	System Design Document; OpenAPI v1.0 frozen
M5	Week of 12 Oct	Mid-Semester Demonstration
M6	Week of 2 Nov	Test Plan and Test Cases
M7	Mon 16 Nov	Final Application, deployed on a public URL (code-complete Sun 15 Nov)
M8	Wed 18 Nov	Final Project Report; presentation, demo and poster from 18 Nov
M9	Thu 19 Nov	Peer Review and Contribution Report

Weekly one-page progress reports run from 27 July to 15 November and are costed inside each member's P1 to P9 hours. Because M7 falls on the first day of P10, the application is **code-complete on 15 November**, the last day of P9; P10 is reserved for the report, presentation, poster, video and peer reviews.

Individual milestones. Each member carries dated milestones with a verifiable exit condition, distinct from the course deadlines above.

Member	Individual milestones with exit conditions
Aum Parmar	OpenAPI v1.0 frozen and signed off — 27 Aug · demo dry run passed end to end — 16 Oct · final report assembled — 18 Nov
Love Shah	Orchestrator routes a two-agent task — 20 Sep · shared memory persists context across sessions — 4 Oct · asynchronous workflow completes end to end — 1 Nov
Panthil Shah	HR and Research agents answer end to end — 4 Oct · six core agents demo-ready — 16 Oct · regression suite green — 8 Nov
Fahim Saiyad	Ingestion indexes a Drive folder — 20 Sep · retrieval returns answers with citations — 4 Oct · seeded base and accuracy report — 8 Nov
Ismail Mansuri	Gmail, Calendar and Slack servers pass integration tests — 20 Sep · GitHub, Jira and Drive with approval hooks — 4 Oct · permission-scope tests green — 8 Nov
Yash Doshi	Authentication and organization UI on staging — 20 Sep · chat workspace usable — 4 Oct · approvals queue demo-ready — 16 Oct · poster delivered — 15 Nov
Param Patel	Authentication service consumable by other services — 13 Sep · approval-queue API live — 11 Oct · security review signed — 15 Nov
Afif Momin	CI green on main — 30 Aug · staging environment complete — 20 Sep · public URL live — 15 Nov

7. Team Task Assignments

Each phase has one accountable lead and one deputy, so every member holds a leadership slot and every module has a named second who can take it over (Section 11). Contributor hours per phase are the columns of Section 4; the named work is in Section 8.

Member	Student ID	Module owned	Deputy for
Aum Parmar	202512004	Project Lead and System Architecture	—
Love Shah	202512058	Agent Orchestration and Shared Memory	Aum Parmar (architecture)
Panthil Shah	202512050	AI Employee Agents	Fahim Saiyad (knowledge base)
Fahim Saiyad	202512103	Knowledge Base and RAG Pipeline	Panthil Shah (agents)
Ismail Mansuri	202512075	MCP Integration Layer	Yash Doshi (frontend)
Yash Doshi	202512094	Frontend (Next.js Dashboard)	Ismail Mansuri (integrations)
Param Patel	202512005	Backend Platform: Gateway, Auth and Organizations	Afif Momin (operations)
Afif Momin	202512063	DevOps, Testing and Documentation	Param Patel (backend)

8. Individual Task Timelines

Each work package decomposes on the Trello board into cards of four to eight hours, approximately 285 in total (per-phase counts in Section 5). A work package is a schedule line, not a task.

Aum Parmar (202512004) — Project Lead and System Architecture — 208 h

Phase	Weeks	Work package	Hours	Due
P1	W1-2	Proposal submission; project plan, effort model and Trello board design	18	9 Aug
P2	W3	BRD consolidation, use-case diagrams, acceptance-criteria review	12	16 Aug
P3	W4-5	Service boundaries, OpenAPI conventions, architecture diagram, design document	26	27 Aug
P4	W6-8	Contract reviews across services, workflow design, integration oversight, sprint facilitation	26	20 Sep
P5	W9-10	Knowledge-base and agent integration oversight, weekly reviews, risk log	22	4 Oct
P6	W11-12	Mid-semester demo scenario, script, dry run and presentation	28	18 Oct
P7	W13-14	Integration oversight, architecture updates, final report drafting begins	24	1 Nov
P8	W15	Test plan review and defect triage coordination	20	8 Nov
P9	W16	Release coordination and security review sign-off	16	15 Nov
P10	W17	Final report assembly, final presentation, peer-review compilation	16	19 Nov

Love Shah (202512058) — Agent Orchestration and Shared Memory — 214 h

Phase	Weeks	Work package	Hours	Due
P1	W1-2	Orchestration scope input; weekly progress reports	8	9 Aug
P2	W3	Orchestration user stories and acceptance criteria	10	16 Aug
P3	W4-5	Orchestrator design, shared-memory schema, sequence diagrams	22	27 Aug
P4	W6-8	Orchestrator skeleton, task routing, agent registry	38	20 Sep
P5	W9-10	Shared memory on PostgreSQL and Redis; agent-to-agent handoffs	34	4 Oct
P6	W11-12	Demo workflow orchestration, approval-gate integration, web integration	28	18 Oct
P7	W13-14	Event-driven asynchronous multi-agent workflows, retries, dead-letter handling	32	1 Nov
P8	W15	Load and failure testing of orchestration paths	16	8 Nov
P9	W16	Orchestrator hardening and deployment support	14	15 Nov
P10	W17	Final report sections, demo support, peer review	12	19 Nov

Panthil Shah (202512050) — AI Employee Agents — 210 h

Phase	Weeks	Work package	Hours	Due
P1	W1-2	Agent scope input; weekly progress reports	8	9 Aug
P2	W3	Agent roles, capabilities and acceptance criteria	10	16 Aug
P3	W4-5	Agent personas, prompt architecture, evaluation plan	22	27 Aug
P4	W6-8	Agent base framework, tool-calling scaffold, prompt templates	26	20 Sep
P5	W9-10	HR and Research agents end to end	36	4 Oct
P6	W11-12	Developer, Engineering Manager, Marketing and Support agents; core set complete	36	18 Oct
P7	W13-14	Agent evaluation harness, prompt tuning, quality benchmarking	32	1 Nov
P8	W15	Agent regression suite	16	8 Nov
P9	W16	Agent configuration documentation and deployment support	12	15 Nov
P10	W17	Final report sections, demo, peer review	12	19 Nov

Fahim Saiyad (202512103) — Knowledge Base and RAG Pipeline — 212 h

Phase	Weeks	Work package	Hours	Due
P1	W1-2	Knowledge base scope input; weekly progress reports	8	9 Aug
P2	W3	Knowledge base requirements and acceptance criteria	10	16 Aug
P3	W4-5	Ingestion pipeline design, chunking and embedding strategy, schema	22	27 Aug
P4	W6-8	Ingestion pipeline, Qdrant vector store, Drive and PDF connectors	38	20 Sep
P5	W9-10	Retrieval with citations; Notion, Confluence, wiki and DOCX connectors	38	4 Oct
P6	W11-12	Retrieval tuning and relevance evaluation for the demonstration	26	18 Oct
P7	W13-14	Retrieval-accuracy harness, re-ranking, chunking optimization	30	1 Nov
P8	W15	Knowledge base seeding with meaningful records; accuracy tests	16	8 Nov
P9	W16	Deployment, index migration, operations runbook	12	15 Nov
P10	W17	Final report sections, demo, peer review	12	19 Nov

Ismail Mansuri (202512075) — MCP Integration Layer — 218 h

Phase	Weeks	Work package	Hours	Due
P1	W1-2	Integration scope input; weekly progress reports	8	9 Aug
P2	W3	Integration requirements and permission model	10	16 Aug
P3	W4-5	MCP permission scope matrix, OAuth flows, approval-hook design	22	27 Aug
P4	W6-8	Gmail, Google Calendar and Slack MCP servers	44	20 Sep
P5	W9-10	GitHub, Jira and Google Drive MCP servers	36	4 Oct
P6	W11-12	Scoped permission enforcement, approval hooks, audit logging, demo integration	32	18 Oct
P7	W13-14	Credential rotation, rate-limit handling, integration resilience	24	1 Nov
P8	W15	Integration test suite, sandbox accounts, permission-scope tests	18	8 Nov
P9	W16	Secret management and integration hardening	12	15 Nov
P10	W17	Final report sections, demo, peer review	12	19 Nov

Yash Doshi (202512094) — Frontend (Next.js Dashboard) — 216 h

Phase	Weeks	Work package	Hours	Due
P1	W1-2	UI scope input; weekly progress reports	8	9 Aug
P2	W3	Initial wireframes and interface requirements	12	16 Aug
P3	W4-5	Near-final Figma mockups, design system, component inventory	22	27 Aug
P4	W6-8	Application shell, authentication pages, organization management UI (4 screens)	34	20 Sep
P5	W9-10	Agent chat workspace; knowledge search UI with citations (3 screens)	32	4 Oct
P6	W11-12	Approvals queue, agent configuration, demonstration dashboards (4 screens)	32	18 Oct
P7	W13-14	Analytics dashboard, notification centre, audit-log viewer (3 screens)	30	1 Nov
P8	W15	UI testing, accessibility and responsive checks	14	8 Nov
P9	W16	UI polish, empty and error states; project poster design	16	15 Nov
P10	W17	Project video walkthrough; final report sections, peer review	16	19 Nov

Param Patel (202512005) — Backend Platform: Gateway, Auth and Organizations — 218 h

Phase	Weeks	Work package	Hours	Due
P1	W1-2	Backend scope input; weekly progress reports	8	9 Aug
P2	W3	Authentication and organization requirements	10	16 Aug
P3	W4-5	ER schema, authentication contracts, RBAC model	18	27 Aug
P4	W6-8	API gateway, authentication service (JWT/OAuth2), organization and user services, RBAC	56	20 Sep
P5	W9-10	Service integration with orchestrator and knowledge base; audit-log service	26	4 Oct
P6	W11-12	Approval queue APIs; notification service	26	18 Oct
P7	W13-14	Analytics APIs, extended endpoints, Postman collection	28	1 Nov
P8	W15	Backend unit and integration test suites	16	8 Nov
P9	W16	Security review, hardening, rate limiting, API documentation	18	15 Nov
P10	W17	Final report sections, demo, peer review	12	19 Nov

Afif Momin (202512063) — DevOps, Testing and Documentation — 216 h

Phase	Weeks	Work package	Hours	Due
P1	W1-2	Git repository, branch strategy, Trello boards, AI prompt log	14	9 Aug
P2	W3	Non-functional and tooling requirements	10	16 Aug
P3	W4-5	CI pipeline, Docker baselines, environment design	20	27 Aug
P4	W6-8	Development and staging environments; thin vertical slice deployed early to de-risk P9	34	20 Sep
P5	W9-10	Pipeline extension for knowledge services; build optimization	24	4 Oct
P6	W11-12	Monitoring, tracing and the demonstration environment	28	18 Oct
P7	W13-14	Autoscaling, log aggregation, test plan authoring, API documentation	32	1 Nov
P8	W15	Test execution coordination, bug log, coverage reporting, user manual	20	8 Nov
P9	W16	Deployment to the target cluster, public URL, operations runbook	20	15 Nov
P10	W17	Peer-review collection, submission packaging, final report sections	14	19 Nov

9. Tools and Task Board

The team uses **Trello**, one of the tools named in the course brief. This supersedes the proposal, which named Jira for sprint tracking; Jira remains in scope only as an MCP integration target inside the product. Two boards are kept in one free workspace.

- **Board A — WBS and Effort.** Ten lists, one per phase, holding every card with its owner, due date, phase label and hours. This is the evidence board and the source of the submitted screenshot.
- **Board B — Sprint Board.** Backlog, Sprint Backlog, In Progress, In Review and Done, on a two-week sprint cadence. Blocked is a red label rather than a list, so a blocked card keeps its true workflow position.

Card titles carry the hours, because a title survives the JSON export and is legible in a screenshot: **[P4][Backend] API gateway routing and rate limiting — Param — est 16h**, with **/ act 14h** appended on completion. The Custom Fields power-up mirrors the same numbers as structured fields. Both boards are set to public visibility and their URLs submitted, so the instructor and teaching assistants have read access without consuming seats against the ten-collaborator limit of a free workspace.

Figure: Sprint Board at submission — Sprint 0 closed, Sprint 1 loaded, later phases held as work packages in the backlog.

Backlog P3-P10 work packages	P3 System Design 29 cards / 174 h	P4 Platform Core 49 cards / 296 h	P5 Knowledge Base & Agents 41 cards / 248 h
	P6 MCP & Demo 39 cards / 236 h	P7 Extended & Hardening 39 cards / 232 h	P8 Testing 23 cards / 136 h
	P9 Deployment 20 cards / 120 h	P10 Final Delivery 18 cards / 106 h	
Sprint Backlog S1 - 14 cards / 84 h	[P2][Backend] Auth + org requirements - Param - est 10h due 16 Aug	[P2][Orchestr] User stories - Love - est 10h due 16 Aug	[P2][Agents] Roles + acceptance criteria - Panthil - est 10h due 16 Aug
	[P2][RAG] KB requirements - Fahim - est 10h due 16 Aug	[P2][MCP] Integration + permission model - Ismail - est 10h due 16 Aug	[P2][Frontend] Wireframes - Yash - est 12h due 16 Aug
	[P2][DevOps] Non-functional + tooling - Afif - est 10h due 16 Aug	[P2][Docs] BRD consolidation - Aum - est 12h due 16 Aug	
In Progress 0 cards	no cards		
In Review 0 cards	no cards		
Done S0 - 13 cards / 80 h	[P1][Docs] Project proposal submitted - Aum - est 10h / act 10h 5 Aug	[P1][Docs] Project plan + effort model - Aum - est 8h / act 8h 9 Aug	[P1][DevOps] Git repo + branch strategy - Afif - est 6h / act 6h 9 Aug
	[P1][DevOps] Trello boards + prompt log - Afif - est 8h / act 8h 9 Aug		

Definition of done. A code card is done when its pull request is merged into main with tests passing in continuous integration, the OpenAPI specification is updated, the change is deployed to staging and the module deputy has reviewed it. A document card is done when the phase lead has reviewed it and it is committed to the repository. A design card is done when it is reviewed in the Friday meeting and exported to the repository.

10. Dependencies and Critical Path

The critical path runs: system design contracts, then the platform core, then the knowledge base and agents, then the integrations and demonstration, then testing, deployment and delivery. Phase P7 carries float because part of its content is stretch scope. The handoffs below are the dates that matter most; a slip on any of them moves the final delivery.

Handoff	From	To	Date
OpenAPI v1.0 frozen	Aum	All service owners	Thu 27 Aug
Authentication service consumable	Param	Yash, Ismail	Sun 13 Sep
Staging environment complete	Afif	All members	Sun 20 Sep
Orchestrator API stable	Love	Panthil	Sun 20 Sep
Retrieval API with citations	Fahim	Panthil, Yash	Sun 4 Oct
Core agents, MCP servers and approvals integrated	Panthil, Ismail, Param	Aum (demo)	Fri 16 Oct
All modules code-complete	All members	Afif (deployment)	Sun 15 Nov

11. Effort Tracking, Reallocation and Evidence

- **Recording.** Estimated hours are fixed at sprint planning. Actual hours are updated by each member at the Friday review, and every logged block references a commit, a pull request, a document revision or minuted meeting, so hours are corroborated rather than asserted.
- **Roll-up.** A committed script parses the weekly Trello JSON export into a planned-versus-actual table per member, stored in the repository, and a burn-up chart against the 1712-hour commitment is included in the final report.
- **Cadence.** Two-week sprints aligned to phase boundaries; planning on Monday, two short stand-ups a week, review and retrospective on Friday. The Friday meeting also produces the weekly progress report.
- **Reallocation.** A member more than 15 per cent behind at a Friday review hands work to the named deputy in Section 7. Two consecutive sprints behind, or an absence beyond one sprint, is raised with the instructor. Released stretch hours are transferred to named committed work — deeper testing, hardening and documentation — never used to reduce a member below 200 hours.

12. Risks, Assumptions and Contingency

Risk	Owner	Likelihood	Trigger	Response
Free cloud tier cannot host Kubernetes, an event bus and three data stores	Afif	High	Credits over 50 per cent consumed by W12	Decided in advance: k3s on a single virtual machine, Redis Streams in place of Kafka for the demonstration, Kubernetes manifests demonstrated locally
Member unavailable or falls behind	Aum	Medium	Two missed stand-ups or 15 per cent variance	Named deputy assumes the module; stretch scope dropped first
Critical-path handoff slips	Aum	Medium	Section 10 date missed by more than three days	Deputy pulled in; stretch dropped; contingency margin of 112 hours applied
Social-platform OAuth approval delayed	Ismail	High	No approval by W11	Those servers are stretch, not committed; demonstration uses the six committed servers
Language-model API spend exceeds budget	Panthil	Medium	80 per cent of the monthly cap	Smaller model for routine calls, cached responses, capped retries
Scope creep	Aum	Medium	New feature requested mid-phase	Added to the stretch backlog only; committed scope changes require re-baselining

Assumptions. Each member sustains approximately 12.6 hours a week for seventeen weeks alongside other courses. The institute academic calendar, including mid-semester examinations and festival holidays falling inside W1 to W17, is overlaid at the first Friday review and the plan re-baselined if any week is lost. Language-model API access is available within a stated monthly budget. All eight members remain on the team for the full semester.

13. Submission Package

Committed to the Git repository and uploaded to Google Classroom for 9 August:

- This plan, containing the timeline chart (Section 5) and the task board (Section 9)
- The timeline chart exported separately from GanttProject, as PDF and PNG
- A screenshot of the live Trello Board A, together with both board URLs
- The Trello import CSV of all work packages and cards, with owners, due dates and hours
- The AI prompt log required by the course brief